5th Year Refraction and Lens Midterm Test

Worked solutions and revision guide

Use this guide to see exactly where marks were won or lost. Read each model answer, compare it with your own copy, then check the common mistakes. Every calculation is shown step by step, because most marks are awarded for the formula, the substitution and the final answer separately.

The marking scheme gives **117 marks** in total. The boxes after each answer show how the marks are shared (thin outline) and the mistakes that cost marks most often (thick outline).

Key formulas and definitions

Every calculation on this test uses one of the relationships below. Learn them, and learn when each one applies.

Idea	Formula or statement	Use it when
Refractive index (Snell's law)	$n = \dfrac{\sin i}{\sin r}$	Light passes between two media and you know two angles.
Refractive index (speeds)	$n = \dfrac{c_{\text{air}}}{c_{\text{glass}}}$	You need the speed of light in a material.
Critical angle (glass to air)	$n = \dfrac{1}{\sin C}$ so $\sin C = \dfrac{1}{n}$	Light travels from glass into air.
Critical angle (core to cladding)	$\sin C = \dfrac{n_{\text{rarer}}}{n_{\text{denser}}}$	Light travels from one glass into another.
Total internal reflection	Light travels in the denser medium and the angle of incidence is greater than C ($i > C$).	Optical fibres and 45°–90°–45° prisms.
Lens formula	$\dfrac{1}{u} + \dfrac{1}{v} = \dfrac{1}{f}$	Real-is-positive convention: u, v and f are positive for real objects, real images and converging lenses.

Golden rule for angles: angles in refraction are always measured from the **normal** (the line at 90° to the surface), never from the surface itself. Several marks on this test were lost by using the wrong one.

Question 1: Optical fibre (28 marks)

An optical fibre has a core (n = 1.47) surrounded by cladding. Glass Y has n = 1.44 and glass Z has n = 1.50.

(i) Which glass, Y or Z, is suitable for the cladding? (4 + 2 marks)

Model answer: Glass Y. Its refractive index (1.44) is lower than the core's (1.47), so light travelling in the core meets a less dense medium at the boundary and can be totally internally reflected.

Marks

- 4 marks: choosing Y.
- 2 marks: the justification, which must say the cladding has a **lower** refractive index than the core, allowing total internal reflection.

Common mistakes

- Choosing Z because it is "closer to the core value" or "stronger". Z has a higher index than the core, so light would refract out into the cladding and the signal would leak away.
- Saying only "Y is lower" without saying lower than **what**.

(ii) Explain what is meant by the critical angle. (4 + 2 marks)

Model answer: The critical angle is the angle of incidence, in the denser medium, for which the angle of refraction in the rarer medium is 90°.

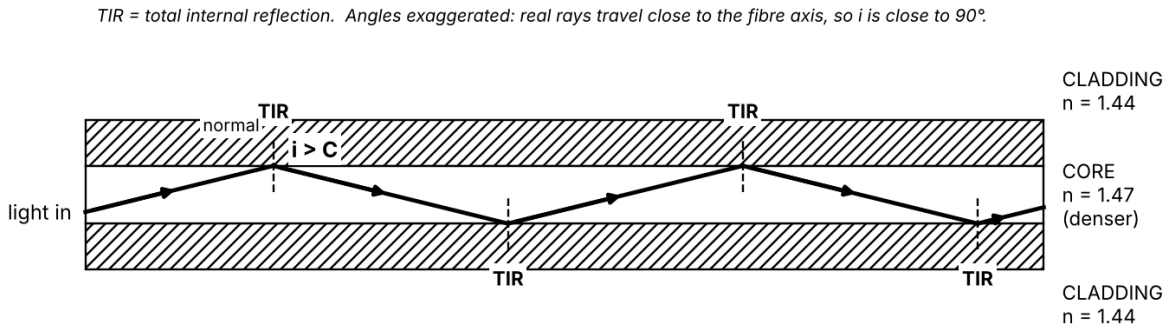
Marks

- 4 marks: angle of incidence in the denser medium.
- 2 marks: the angle of refraction is 90° (the refracted ray travels along the boundary).

Common mistakes

- Writing "the angle at which light bends" or "the angle where total internal reflection happens". Total internal reflection happens at angles **greater than** the critical angle, not at it.
- Forgetting to say the light must be going from the denser medium to the rarer one.

(iii) With a labelled diagram, explain how a signal is transmitted along the fibre. (4 + 2 marks)



Light travelling along an optical fibre by repeated total internal reflection

Model answer: Light enters one end of the core and strikes the core–cladding boundary. The angle of incidence is greater than the critical angle (i > C), so the light is totally internally reflected. It zigzags along the core by repeated total internal reflection until it leaves at the other end, carrying the signal with it.

Marks

- 4 marks: a clear diagram showing the light ray reflecting inside the fibre.
- 2 marks: stating i > C or total internal reflection.

Common mistakes

- A diagram with no labels, or a ray that simply passes straight through.
- Drawing the ray refracting out of the core.
- Drawing reflections with unequal angles: the angle of incidence must equal the angle of reflection.
- Forgetting the statement i > C.

(iv) Calculate the critical angle between the core and the cladding. (3 + 3 marks)

Light goes from the core (denser, n = 1.47) into the cladding (rarer, n = 1.44), so use the ratio of the two indices.

$$\sin C = \frac{n_{\text{cladding}}}{n_{\text{core}}} = \frac{1.44}{1.47} = 0.9796$$

$$C = \sin^{-1}(0.9796) = 78.4^\circ$$

Answer: C = 78.4°

Marks

- 3 marks: the correct formula (n = sin i ÷ sin r with r = 90°, or n = 1 ÷ sin C).
- 3 marks: C = 78.4°.

Common mistakes

- Using sin C = 1 ÷ 1.47, which gives 42.9°. That is the critical angle between the core and **air**, not the cladding.
- Turning the fraction upside down. 1.47 ÷ 1.44 is greater than 1, and sin C can never be greater than 1.
- A calculator left in radian mode.

(v) Which glass, from the table, lets light travel fastest? (4 marks)

Model answer: Glass Y. Since n = c(air) ÷ c(glass), the glass with the **lowest** refractive index has the highest speed of light.

Glass	Refractive index, n	Speed = $3.0 \times 10^8 \div$ n (m/s)
Y	1.44	2.08×10^8
Core	1.47	2.04×10^8
Z	1.50	2.00×10^8

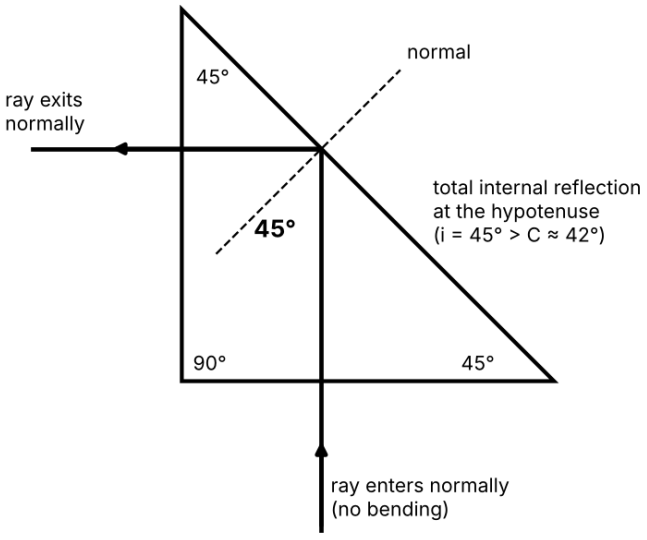
The scheme accepts $n = c_1 \div c_2$ for 3 marks if the glass is not identified correctly.

Common mistakes

- Choosing Z because it has the biggest number. A larger refractive index means light is slowed **more**.

Question 2: Turning light through 90° with a prism (7 marks)

Task: Draw a diagram to show how a ray of light can be turned through 90° using a 45°–90°–45° prism.



A ray entering a 45°–90°–45° prism normally, reflecting totally at the hypotenuse and leaving normally

How to build the diagram, step by step:

1. Draw the right-angled triangle with the long side (the hypotenuse) facing the direction you want the light to turn.
2. Draw the ray hitting one **short** face at 90° (along the normal). It does not bend, because the angle of incidence is 0°.
3. The ray reaches the hypotenuse at an angle of incidence of 45°. The critical angle for glass is about 42° (for n = 1.5), and 45° is greater than that, so the ray is **totally internally reflected**.
4. The reflected ray travels at 90° to its original direction and hits the other short face along the normal, so it leaves without bending.

Marks

- 2 marks: ray entering normally at one short face.
- 2 marks: ray exiting at the other short face.
- 3 marks: total internal reflection at the hypotenuse.

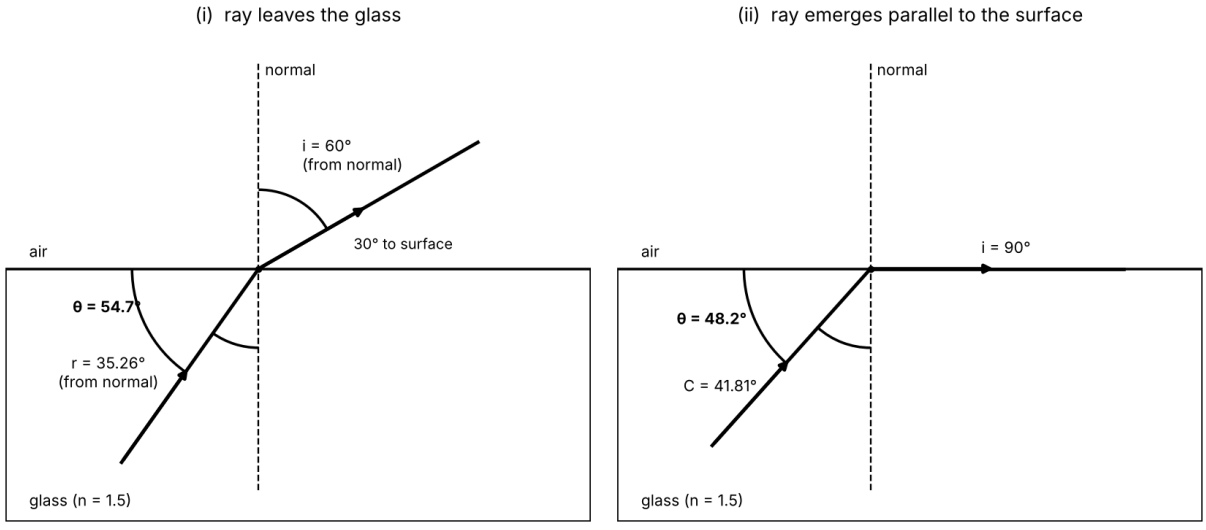
Common mistakes

- Letting the ray enter through the hypotenuse, so the 90° turn does not happen.
- Drawing the ray bending at the short faces when it enters along the normal.
- Showing the ray reflecting but never saying why. Label it total internal reflection (TIR) and write i = 45° > C.
- Measuring the angle at the hypotenuse to the surface instead of to the normal.

Question 3: Ray leaving a glass block (28 marks)

A ray leaves a rectangular glass block (n = 1.5). Inside the glass it makes an angle θ with the surface. In the air it makes an angle of 30° with the surface.

The key idea for this whole question: refraction angles are measured from the **normal**, not from the surface. The question gives angles to the surface, so convert first: angle to normal = 90° − angle to surface.



Geometry for Question 3, parts (i) and (ii)

(i) Calculate θ when n = 1.5. (12 marks: 3 + 3 + 3 + 3)

Step 1: formula (3 marks).

$$n = \frac{\sin i}{\sin r}$$

Step 2: angle in air, from the normal (3 marks). The ray is 30° from the surface, so:

$$i = 90^\circ - 30^\circ = 60^\circ$$

Step 3: angle in glass, from the normal (3 marks).

$$\sin r = \frac{\sin 60^\circ}{1.5} = \frac{0.8660}{1.5} = 0.5774$$

$$r = \sin^{-1}(0.5774) = 35.26^\circ$$

Step 4: convert back to the angle with the surface (3 marks).

$$\theta = 90^\circ - 35.26^\circ = 54.7^\circ$$

Answer: θ = 54.7°

Common mistakes

- Using 30° directly as the angle in air.
- Stopping at r = 35.26° and writing it as θ, when θ is measured from the surface.
- Putting the formula the wrong way up. sin r = 1.5 × sin 60° = 1.30 is impossible, and that is the clue you have inverted it.
- The angle in air is the larger one, because light bends **away** from the normal when it goes from glass into air.

(ii) What value of θ makes the ray emerge parallel to the side of the block? (9 marks: 3 + 3 + 3)

Idea (3 marks): a ray emerging parallel to the surface travels along the boundary, so its angle of refraction in air is 90°. The angle in the glass is then the **critical angle**.

$$\sin i_c = \frac{1}{n} = \frac{1}{1.5} = 0.6667$$

Critical angle (3 marks):

$$i_c = \sin^{-1}(0.6667) = 41.81^\circ$$

Convert to the angle with the surface (3 marks):

$$\theta = 90^\circ - 41.81^\circ = 48.2^\circ$$

Answer: θ = 48.2°

Common mistakes

- Writing θ = 41.8° and forgetting that θ is measured from the surface, not from the normal.
- Not mentioning the critical angle at all, which loses the first 3 marks even if the later numbers are right.
- Finding sin of the critical angle as 1.5 instead of 1 ÷ 1.5. The sine of an angle can never be more than 1.

(iii) Calculate the speed of light in the glass. (7 marks: 4 + 3)

$$c_{\text{glass}} = \frac{c_{\text{air}}}{n} = \frac{3.0 \times 10^8}{1.5} = 2.0 \times 10^8 \text{ m/s}$$

Answer: c(glass) = 2.0 × 10^8 m/s

Marks

- 4 marks: the formula.
- 3 marks: the answer with correct units. One mark is deducted for missing or wrong units.

The marking scheme uses 2.9979 × 10^8 m/s for the speed of light in air, which also rounds to 2 × 10^8 m/s.

Common mistakes

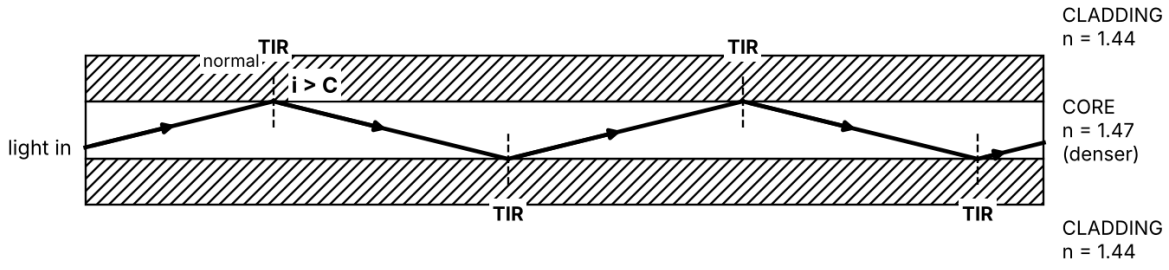
- Multiplying by 1.5 instead of dividing, giving 4.5 × 10^8 m/s, which is faster than light in air. Light is **slower** in glass, so your answer must be smaller than 3.0 × 10^8 m/s.
- Leaving out the units (m/s).

Question 4: Guiding light along an optical fibre (28 marks)

The fibre has a core of high quality glass with n = 1.55, coated with glass of lower refractive index. Light must travel through a very long length of fibre.

(a) Explain, with a labelled diagram, how a ray of light is guided along a fibre. (9 marks: 3 + 3 + 3)

TIR = total internal reflection. Angles exaggerated: real rays travel close to the fibre axis, so i is close to 90°.



Light guided along a fibre by total internal reflection

Model answer: The ray enters the core and hits the boundary with the cladding at an angle of incidence greater than the critical angle. Because the light is going from the denser core to the rarer cladding, it is totally internally reflected. It then reflects again and again along the fibre until it reaches the far end.

Marks

- 3 marks: a diagram showing a light ray inside a glass fibre.
- 3 marks: the ray reflected at least once.
- 3 marks: a reference to the critical angle or total internal reflection.
- 1 mark is deducted if the diagram has no labels. Label the core, the cladding, the normal and the angles.

Common mistakes

- An unlabelled diagram.
- A ray that leaves the fibre instead of reflecting.
- Drawing the fibre but never naming the process. Write "total internal reflection" and "i > critical angle" on your answer.

(b) Why is each fibre coated with glass of lower refractive index? (6 marks: 3 + 3)

Model answer: So that the light is going from a denser medium (the core) to a rarer medium (the cladding), which is the condition for total internal reflection. With the lower-index coating the angle of incidence can be greater than the critical angle, so all the light is reflected back into the core and none escapes.

Marks

- 3 marks: light travels from the denser to the rarer medium.
- 3 marks: total internal reflection occurs (i > C), so no light escapes.

Common mistakes

- Saying the coating "protects the glass" or "stops it breaking". It may do that too, but it is not the physics reason the question wants.
- Saying "for reflection" without saying **total internal** reflection.

(c) What is the speed of light in the fibre? (7 marks: 4 + 3)

$$c_{\text{glass}} = \frac{c_{\text{air}}}{n} = \frac{3.0 \times 10^8}{1.55} = 1.94 \times 10^8 \text{ m/s}$$

Answer: c(glass) = 1.94×10^8 m/s

Marks

- 4 marks: the formula and substitution.
- 3 marks: the answer. The scheme gives no penalty for units on this part, but always include them (m/s).

Common mistakes

- Multiplying instead of dividing. 4.65×10^8 m/s is faster than light in air, which is impossible.
- Using the refractive index of the cladding instead of the core.

(d) What power is transmitted after 8 km? (6 marks: 3 + 3)

The power halves every 2 km. Over 8 km that is 8 ÷ 2 = 4 halvings.

$$P = 10 \times \left(\frac{1}{2}\right)^4 = \frac{10}{16} = 0.625 \text{ W}$$

Answer: P = 0.625 W (or 5/8 W)

Distance travelled	Power remaining
0 km	10 W
2 km	5 W
4 km	2.5 W
6 km	1.25 W
8 km	0.625 W

Marks

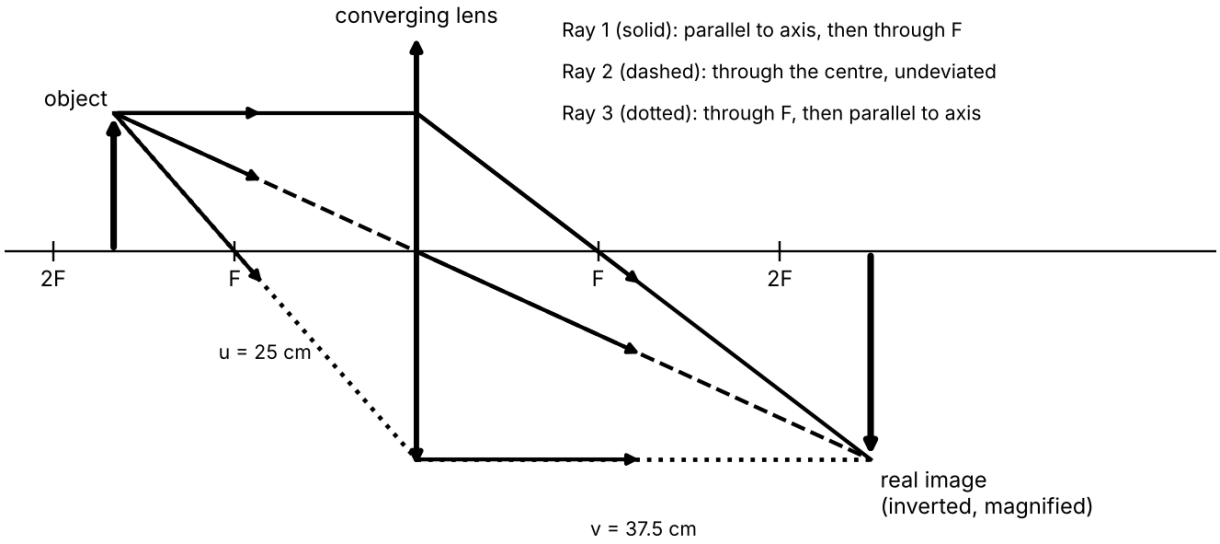
- 3 marks: the method, $(½)^4$, or 1/16 of the original, or 10 ÷ 16.
- 3 marks: the answer, 0.625 W (or 5/8 W).

Common mistakes

- Subtracting instead of halving each time (10 − 8 = 2).
- Using only one halving (5 W), or dividing by 8 (1.25 W).
- Using $(½)^8$ because the distance is 8 km. The loss happens every **2 km**, so count the 2 km steps.

Question 5: Ray diagram for a real image (12 marks)

Task: Draw a converging lens and a ray diagram to show how the lens forms a real image.



An object between F and 2F forms a real, inverted, magnified image beyond 2F

This diagram is drawn to scale using the numbers from Question 6 (f = 15 cm, object 25 cm from the lens), so it gives the same image position you calculate there.

How to draw it:

1. Draw the principal axis and a converging lens (a vertical line with outward arrowheads). Mark the focal points F on both sides, and 2F if you can.
2. Place the object as an upright arrow **beyond F** on the left. A real image needs the object further from the lens than the focal length.
3. Draw at least two of these three rays from the top of the object:
   - Parallel to the axis, then refracted through the far focal point F.
   - Through the optical centre of the lens, which carries straight on.
   - Through the near focal point F, which emerges parallel to the axis.
4. Where the rays meet, draw the image as an arrow. It is **real** because rays actually cross there, so it can be shown on a screen.

Marks

- 3 marks: a converging lens drawn correctly.
- 3 marks: the object shown.
- 3 marks: the incident ray or rays drawn correctly.
- 3 marks: the image shown where the rays meet.

Common mistakes

- Drawing the lens as a diverging lens (with inward arrowheads).
- Placing the object **inside** F, where the image is virtual and the refracted rays spread apart.
- Rays that do not meet at a point, or rays with no arrows.
- Drawing the image upright. A real image from a converging lens is inverted.
- Forgetting to mark F, so the rays cannot be checked.

Question 6: Lens formula calculation (9 marks)

Task: An object is 25 cm in front of a converging lens of focal length 15 cm. Calculate the position of the image.

Step 1: formula (3 marks).

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

Step 2: substitution (3 marks). Rearrange to make 1/v the subject.

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{15} - \frac{1}{25} = \frac{5-3}{75} = \frac{2}{75}$$

Step 3: answer (3 marks). Turn the fraction over.

$$v = \frac{75}{2} = 37.5 \text{ cm}$$

Answer: v = 37.5 cm, on the opposite side of the lens from the object

The image is real, inverted and magnified. The magnification is m = v ÷ u = 37.5 ÷ 25 = 1.5.

The scheme allows 3 marks for a partial answer, for example a correct value of 1/v that has not been inverted.

Common mistakes

- Adding instead of subtracting. 1/15 + 1/25 gives 9.4 cm. Check: an object between F and 2F must give an image beyond 2F (further than 30 cm), so 9.4 cm cannot be right.
- Stopping at 1/v = 0.0267 and forgetting to take the reciprocal.
- Putting 15 and 25 in the wrong places in the formula.
- Leaving out the unit (cm).

Question 7: Sight defect corrected by a converging lens (5 marks)

Model answer: Long-sightedness, also called hypermetropia or hyperopia. Any of these three names earns full marks.

Why: in a long-sighted eye the eyeball is too short, or the eye lens is too weak, so light from a close object would come to a focus **behind** the retina. The person sees distant objects clearly but near objects are blurred. A converging lens in the glasses bends the light inwards before it enters the eye, so the image forms on the retina.

The scheme allows 3 marks for a partial answer, such as describing the symptom (difficulty seeing near objects) without naming the defect.

Defect	Also called	Problem	Corrected with
Long-sightedness	Hypermetropia (hyperopia)	Near objects blurred; image forms behind the retina	Converging (convex) lens
Short-sightedness	Myopia	Distant objects blurred; image forms in front of the retina	Diverging (concave) lens

Common mistakes

- Writing short-sightedness (myopia), which is corrected by a **diverging** lens.
- Mixing up the names. Remember: hypermetropia = long-sighted = converging lens.

Marks summary and revision checklist

Fill in your own marks and note the main fix for each question.

Q	Topic	Marks available	My marks	Main thing to fix
1	Optical fibre: cladding, critical angle, TIR	28		
2	45°–90°–45° prism	7		
3	Ray leaving a glass block	28		
4	Guiding light along a fibre, speed, power loss	28		
5	Ray diagram, real image	12		
6	Lens formula	9		
7	Sight defect	5		
Total		**117**		

Before the next test, can you do each of these without notes?

- ☐ Define the critical angle in one sentence, including "denser medium" and "90°".
- ☐ State the two conditions for total internal reflection.
- ☐ Choose the cladding glass and say why its refractive index must be lower than the core's.
- ☐ Convert an angle measured from the surface into an angle from the normal.
- ☐ Use n = sin i ÷ sin r, n = 1 ÷ sin C and n = c(air) ÷ c(glass) in the correct situations.
- ☐ Check that an answer is sensible: light is slower in glass, and the sine of any angle is never more than 1.
- ☐ Draw a labelled fibre diagram and a 45°–90°–45° prism diagram with total internal reflection marked.
- ☐ Draw a converging-lens ray diagram with three rays and an object beyond F.
- ☐ Rearrange 1/u + 1/v = 1/f to find v, and remember to take the reciprocal at the end.
- ☐ Name the defect corrected by a converging lens and the one corrected by a diverging lens.

Exam habits that protect marks: write the formula first, then substitute, then give the answer with units. Label every diagram. Use the key words the scheme rewards: critical angle, total internal reflection, denser to rarer.